

JESSICA QIU

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EDUCATION

Boston University College of Engineering

May 2027

Bachelor of Science in Electrical Engineering, Minor in Biomedical Engineering

GPA: 3.8

Relevant Coursework: Electrical Circuits, Logic Design, Signals and Systems, Electromagnetic Systems, Power Electronics & Energy Systems, Digital Signal Processing.

Activities & Societies: Theta Tau Professional Engineering Fraternity (Academic & Professional Development Chair); BU Mars Rover Club (Electrical & Life Detection Team); BU Society of Asian Scientists and Engineers (Mentor).

EXPERIENCE

Undergraduate Research Assistant

May 2026 – July 2026

Centre for Bio-Devices and Bio-Informatics, Nanyang Technological University

Singapore

- Engineered and fabricated flexible PMMA-based optical sensors, optimizing polymer molecular weight and fabrication parameters to improve fiber stability, mechanical durability, and sensing performance for biomedical applications.
- Performed mechanical and optical characterization of microlaser sensors under applied stress using microscopy, laser excitation, and spectroscopy to quantify stress-induced spectral changes and evaluate sensor performance.
- Achieved 0.054 nm/MPa sensitivity, $R^2 = 0.99$, and 3.03 MPa limit of detection, demonstrating an optical sensing platform for finger-joint motion monitoring and hand mobility assessment through stress-induced microlaser responses.

Undergraduate Research Assistant

Nov 2024 – Present

BU Ji-Xin Cheng Research Group

Boston, MA

- Engineered optical beam paths for VREF and SRP nonlinear imaging systems, integrating a Carbide Yb laser (1030/800/1500 nm), polarization optics, delay stages, and PMTs to optimize mid-IR excitation and system stability.
- Quantified photothermal responses and DFG output efficiency using spectroscopic datasets, achieving a 5.7 signal-to-background ratio (SBR) and 20 cm^{-1} spectral resolution (FWHM) in C–D silent-region spectra at 2125 cm^{-1} .
- Collaborated closely with postdocs and biologists to perform quantitative image analysis on *C. elegans* models, utilizing fluorescence microscopy to map lysosome-related organelles and investigate cellular stress response.

Undergraduate Teaching Assistant - ENG EC 410: Intro to Electronics

Sept 2026 – Present

BU College of Engineering

Boston, MA

- Conducted weekly electronics labs covering diode, BJT, and MOSFET circuits, providing hands-on instruction in circuit design, breadboarding, and practical measurement techniques using oscilloscopes and multimeters.
- Evaluated and graded homework, circuit design reports, and lab assignments, providing detailed feedback on circuit analysis, SPICE/LTspice simulations, and troubleshooting to reinforce small-signal and frequency-domain concepts.

Undergraduate Teaching Assistant - ENG BE/EC 556: Optical Spectroscopic Imaging

Jan 2025 – Present

BU College of Engineering

Boston, MA

- Led weekly office hours and provided support for graduate-level coursework in optical spectroscopy and imaging, guiding students through laser physics, wave optics, and spectroscopic analysis to strengthen problem-solving skills.
- Evaluated assignments, exams, and projects, providing detailed feedback to reinforce key concepts in spectroscopic imaging and microscopy, while collaborating with the instructor on course administration and instructional support.

Software Engineering Intern I

June 2024 – August 2025

Pulp Internet Corporation

Brooklyn, NY

- Developed interactive Data Stories templates and automated data pipelines using JavaScript, Python, pandas, NumPy, and Streamlit, transforming transcript datasets into interactive visualizations and narratives.
- Contributed to Persona Engine development by integrating OpenAI workflows and NLP techniques to analyze communication patterns, extract key themes, and generate audience-specific insights from large transcript datasets.

PROJECTS

Biometric Sensor for Exercise Device

Oct 2024 – Dec 2024

- Led a 4-person team in the design of a wearable biometric sensing device that monitored wrist acceleration and rotation during tennis swings, providing real-time haptic feedback when user-defined performance thresholds were met.
- Designed and integrated the device's electronic hardware, including sensor, power, and haptic-feedback circuitry, using an Arduino Nano, IMU, OLED display, and rotary encoder to enable real-time motion tracking and user interaction.
- Developed firmware to process acceleration and gyroscope data, support configurable threshold settings, and achieve a $\pm 5\%$ measurement accuracy target through calibration, testing, and iterative hardware-software refinement.

SKILLS

Hardware: Oscilloscopes, Function Generators, PCB Design, Soldering, Microcontrollers, Optical Alignment, Fluorescence Microscopy, VREF & SRP Microscopy, Mid-IR Photothermal (MIPT) Imaging, Yb Lasers, PMTs, Polarization Optics

Software: KiCad, LaTeX, Microsoft Office, Revit, AutoCAD, Python, MATLAB, Verilog, JavaScript.